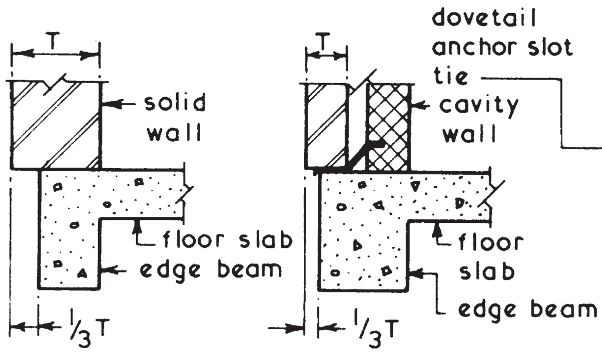
Non-load-bearing Brick Panel Walls ~ these are used in conjunction with framed structures as an infill between the beams and columns. They are constructed in the same manner as ordinary brick walls with the openings being formed by traditional methods.

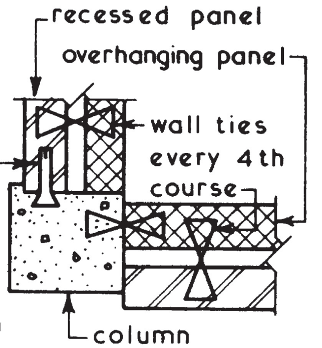
Basic requirements:

1. To be adequately supported by and tied to the structural frame.
2. Have sufficient strength to support own self-weight plus any attached finishes and imposed loads such as wind pressures.
3. Provide the necessary resistance to penetration by the natural elements.
4. Provide the required degree of thermal insulation, sound insulation and fire resistance.
5. Have sufficient durability to reduce maintenance costs to a minimum.
6. Provide for movements due to moisture and thermal expansion of the panel and for contraction of the frame.

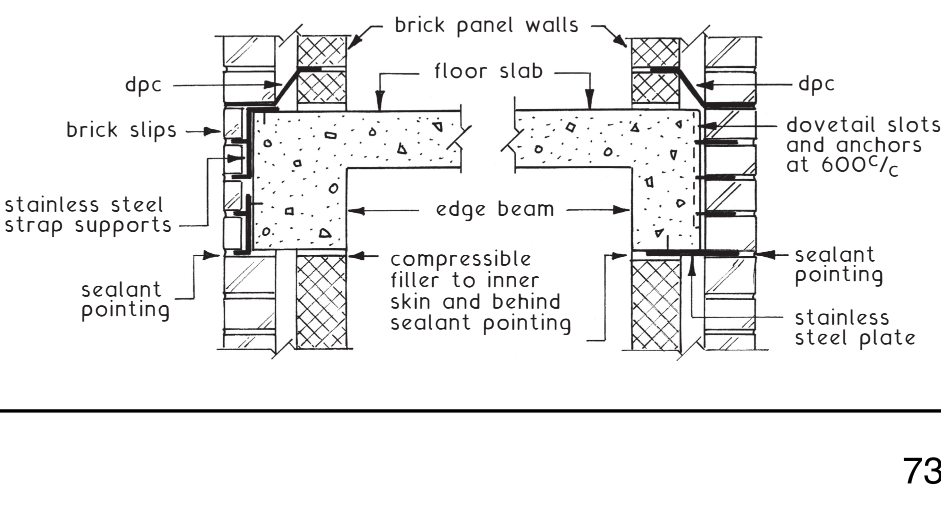
Typical Details ~

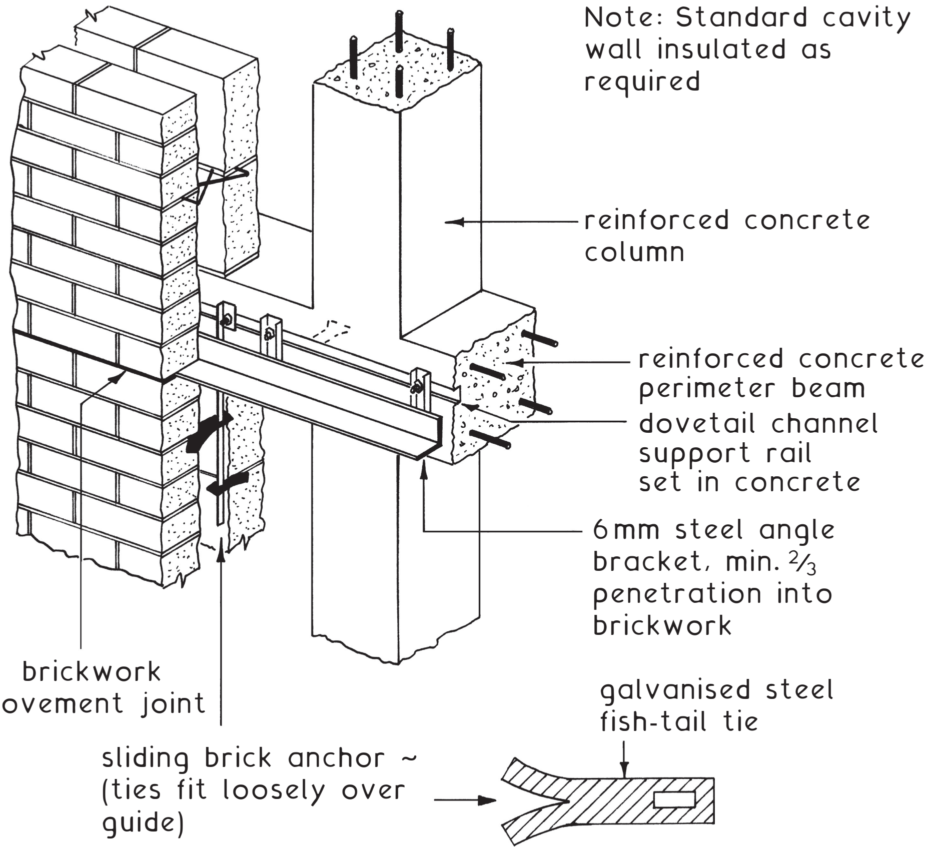


MAXIMUM PERMISSIBLE OVERHANG



COLUMN ABUTMENTS



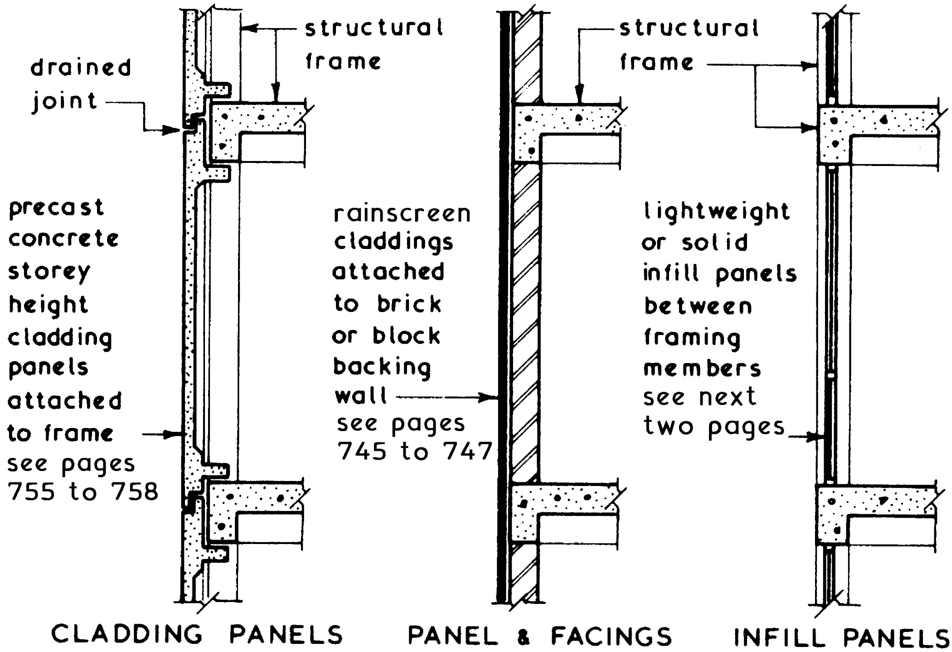


Application – multi-storey buildings, where a traditional brick façade is required.

Brickwork movement – to allow for climatic changes and differential movement between the cladding and main structure, a 'soft' joint (cellular polyethylene, cellular polyurethane, expanded rubber or sponge rubber with polysulphide or silicon pointing) should be located below the support angle. Vertical movement joints may also be required at a maximum of 12 m spacing.

Lateral restraint – provided by normal wall ties between inner and outer leaf of masonry, plus sliding brick anchors below the support angle.

Infill Panel Walls ~ these can be used between the framing members of a building to provide the cladding and division between the internal and external environments and are distinct from claddings and facing:

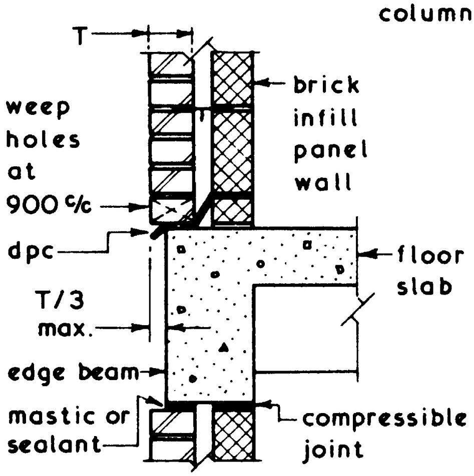


Functional Requirements ~ all forms of infill panel should be designed and constructed to fulfil the following functional requirements:

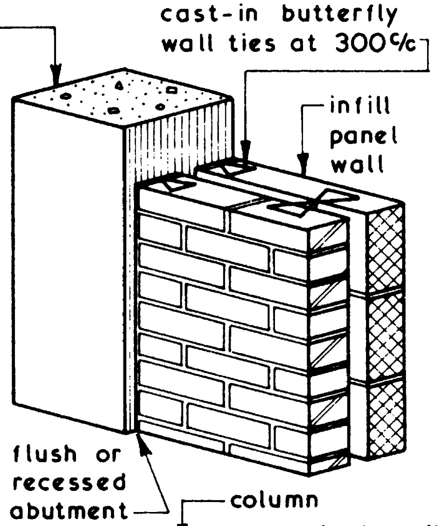
1. Self-supporting between structural framing members.
2. Provide resistance to the penetration of the elements.
3. Provide resistance to positive and negative wind pressures.
4. Give the required degree of thermal insulation.
5. Give the required degree of sound insulation.
6. Give the required degree of fire resistance.
7. Have sufficient openings to provide the required amount of natural ventilation.
8. Have sufficient glazed area to fulfil the natural daylight and vision out requirements.
9. Be economical in the context of construction and maintenance.
10. Provide for any differential movements between panel and structural frame.

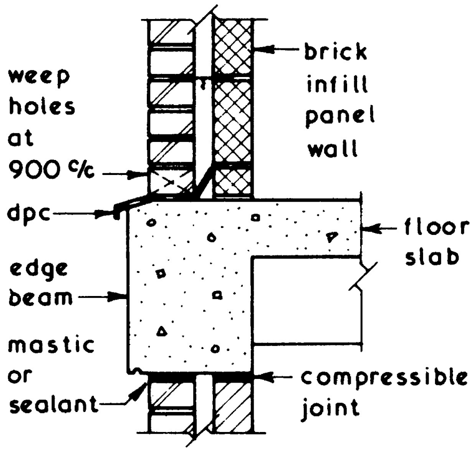
Brick Infill Panels ~ these can be constructed in a solid or cavity format, the latter usually having an inner skin of blockwork to increase the thermal insulation properties of the panel. All the fundamental construction processes and detail of solid and cavity walls (bonding, lintels over openings, wall ties, damp-proof courses, etc.) apply equally to infill panel walls. The infill panel walls can be tied to the columns by means of wall ties cast into the columns at 300 mm centres or located in cast-in dovetail anchor slots. The head of every infill panel should have a compressible joint to allow for any differential movements between the frame and panel.

Typical Details

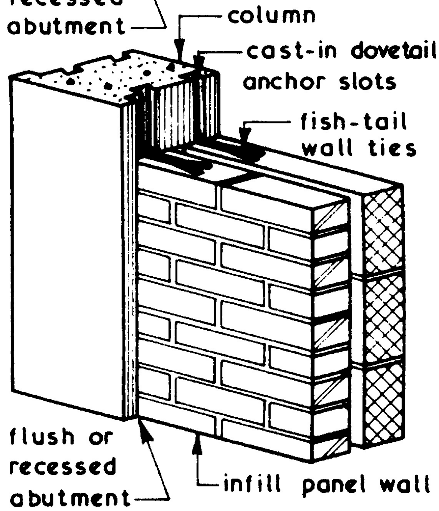


PROJECTING INFILL PANEL



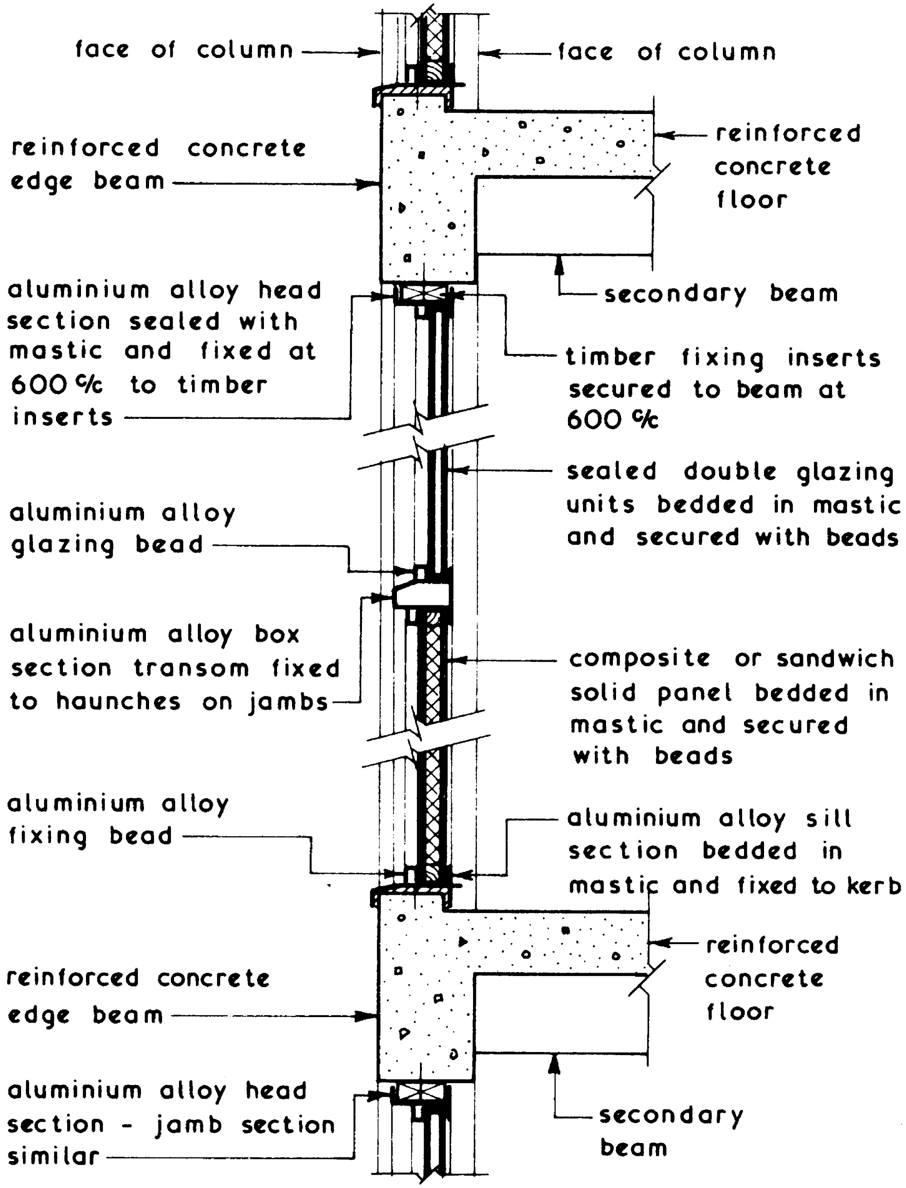


RECESSED INFILL PANEL

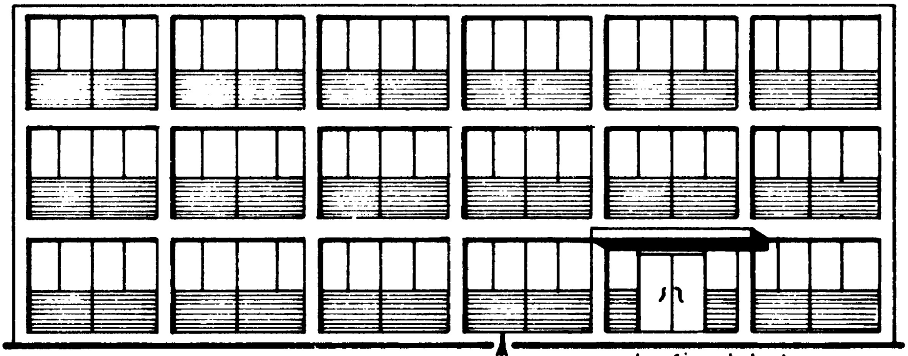


Lightweight Infill Panels ~ these can be constructed from a wide variety or combination of materials such as timber, metals and plastics into which single or double glazing can be fitted. If solid panels are to be used below a transom they are usually of a composite or sandwich construction to provide the required sound insulation, thermal insulation and fire resistance properties.

Typical Example ~

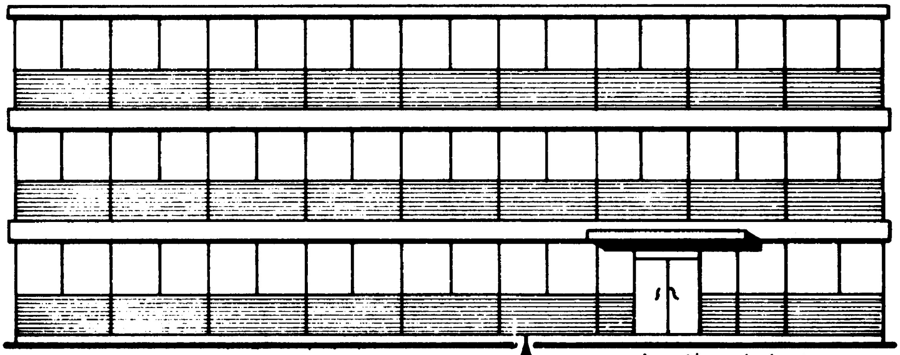


Lightweight Infill Panels ~ these can be fixed between the structural horizontal and vertical members of the frame or fixed to the face of either the columns or beams to give a grid, horizontal or vertical emphasis to the façade thus:



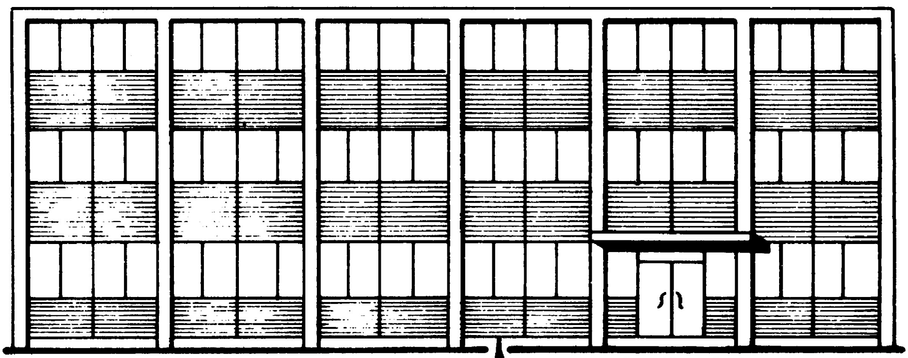
GRID OR FRAME EMPHASIS

panels fixed between columns and beams



HORIZONTAL EMPHASIS

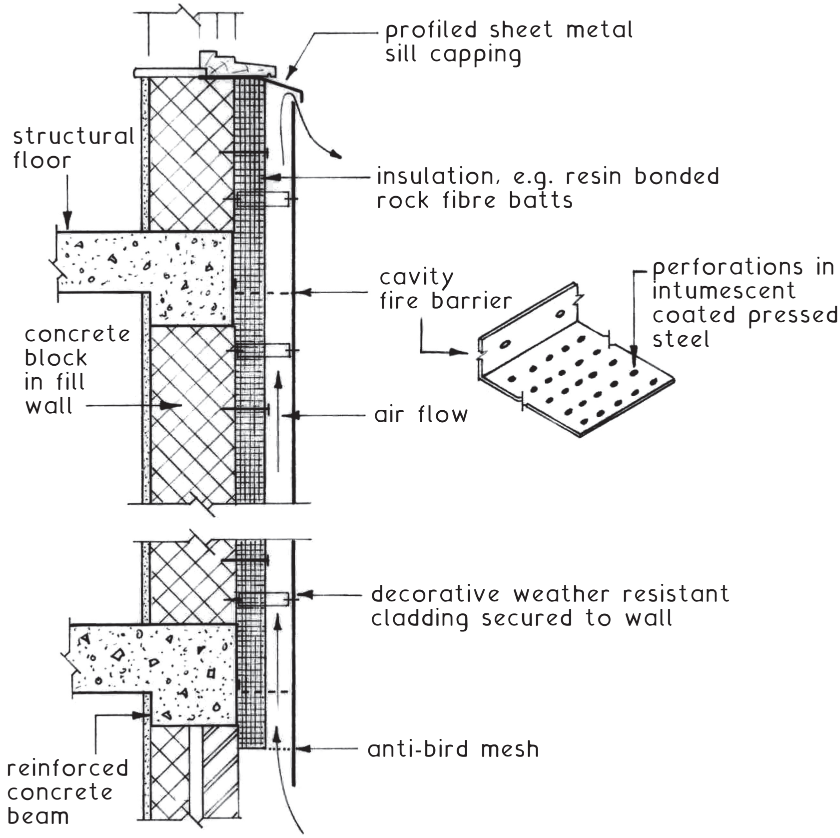
panels fixed between beams and over columns



VERTICAL EMPHASIS

panels fixed between columns and over beams

Rainscreen Cladding ~ a surface finish produced by overcladding the external walls of new construction, or as a decorative façade and insulation enhancement to the external walls of existing construction. This concept provides an inexpensive *loose-fit* weather resistant layer. It is simple to replace to suit changes in occupancy, corporate image, client tastes, new material innovations and design changes in the appearance of buildings. Sustainability objectives are satisfied by re-use and refurbishment instead of demolition and rebuilding. Existing buildings can be seamlessly extended with uniformity of finish.



Principal Features ~

* Weather-proof outer layer includes plastic laminates, fibre-cement, ceramics, aluminium, enamelled steel and various stone effects.
* Decorative finish.
* Support frame attached to structural wall.
* Ventilated and drained cavity behind cladding.
* Facility for sound and thermal insulation.
* *Loose-fit* simple and economic for ease of replacement.